UNIVERSITY OF TECHNOLOGY FACULTY OF CSE	FINAL EXAM	Semester / Academic year	2	2022-2023
		Date		30/05/2023
	Course title	Discrete Structure for Computing		
	Course ID	CO1007		
	Duration	80 mins	Question sheet code	2225

Notes: - Students do not use course materials except one A4 hand-writing document.
- Submit the question sheet together with the answer sheet.
- Choose the best answer (only 1) for each question.

1. (L.O.2.2) What is the probability that

$$x + y + z \equiv 1 \mod p,$$

when x, y and z are chosen uniformly at random from $\{0, \ldots, p-1\}$ with an odd prime number p?

A. $1/3p$. B. $1/p^3$. C. $3/p$. D. $1/p$.

For questions 2–6, consider the following adjacency matrix for the simple undirected graph H (the weight 0 means no connection):

	A	B	C	D	E	F	G
A	0	8	4	1	0	2	0
B	8	0	0	0	5	0	2
C	4	0	0	9	0	1	3
D	1	0	9	0	4	0	2
E	0	5	0	4	0	7	1
F	2	0	1	0	7	0	6
G	0	2	3	2	1	6	0

2. (L.O.2.2) Assume that the index of the cell (or the graph's edge) in the adjacency matrix will be indexed by order of alphabet and number (from 1 to the end). For instance, cell (A, B) is indexed $(1, 2)$, cell (A, A) is $(1, 1)$, cell (A, C) is $(1, 3)$, ..., (G, G) is $(7, 7)$.
Start from A; the initialize step is Step 0; find the value of the Bellman-Ford algorithm's Third step (Step 3). Note that the weight of an edge is not fixed; instead, it will be changed following the rule:
The weight of the cell/edge where the sum of the row and column index is an odd number will be reduced by 1 after every 2 step of calculation (if the weight is 1, it cannot reduce more). For instance, at the first and second steps, the value of cell $(1, 2)$ a.k.a edge (A, B) is 8, and when calculating at the third and fourth steps, it will be 7 and will be further decreased to 6 when calculating fifth and sixth, ...

A. 0,3,2,1,3,1,2 B. 0,4,3,1,4,1,2 C. 0,8,3,1,5,2,3 D. 0,5,3,1,4,2,3

3. (L.O.2.2) Can we remove an edge from H to create H_1 so that H_1 has an Euler circuit? If yes, what edge can we remove?

A. Cannot B. Can, remove edge BG
C. All the other choices are incorrect D. Can, remove edge AF

4. (L.O.2.2) Apply BFS from B to find a spanning tree of H. What is the total weight of this spanning tree? Note: When choosing between multiple vertices, choose according to the alphabet in ascending order.

A. 16 B. 22 C. 12 D. 9

5. (L.O.2.2) Find the minimum spanning tree of H with the root D, we follow the criteria:

   - If a child is earlier than its parent in the alphabet, it is a left child, else it is a right child
   - There can be more than 1 left child or more than 1 right child.
   - When choosing between multiple edges, select the edge with the earliest alphabet in ascending order of the alphabet list.

   What is the in-order traverse of the obtained tree?

   A. F C A D G B E B. A C F D B E G C. C F A B G E D D. C F A D B G E

6. (L.O.2.2) Given a set X consisting of sub-graphs S with 6 edges and a vertex A of degree 4. What is the probability of choosing a graph from X and that is a spanning tree? know that the probability of choosing the vertices is the same

   A. 0.4444 B. 0.4222 C. 0.0066 D. 0.0063

7. (L.O.3.1) Let $G = (V, E)$ be a weighted directed graph with $V = \{1, ..., 100\}$. The edges of the graph and its weights $w(i, j)$ are assigned according to the following rule:

$$E = \{(i, j) | j = i + 1, \text{or } j = 3i\},\ w(i, j) = 1,$$

   for all $(i, j) \in E$. The shortest path in G from vertex $u = 1$ to vertex $v = 100$ has length

   A. 33. B. 4. C. 27. D. 7.

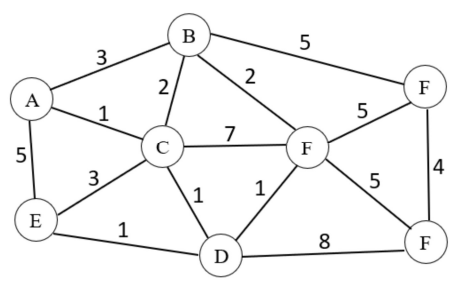
8. (L.O.1.2) How many arrangements of the letters in $ABRACADABRA$ have no A adjacent to a B?

   A. 5880 B. 5660 C. 5520 D. 3360

9. (L.O.1.2) How many ways are there to select **five bills** from a cash box containing \$1, \$2, \$5, \$10, \$20, \$50, and \$100 bills? Assume that the order in which the bills are chosen does not matter; the bills of each denomination are indistinguishable, and there are at least five bills of each type.

   A. 1287 B. 462 C. 330 D. 792

10. (L.O.3.2) Consider the weighted undirected graph G



   What is the total weight of the minimum spanning tree rooted at A obtained using Prim's algorithm?

   A. 30 B. 18 C. 12 D. 15

11. (L.O.3.1) Assume the population's disease B ratio is 1/10000. What is the probability of a person having B, given that person tests positive for B? Assume that there is a 99.9% chance a person with the disease tests positive, and a 0.02% chance a person without B tests positive for it. Choose the best approximate of the probability.

   A. 52.25% B. 33.31% C. 75.53% D. 18.02%

12. (L.O.3.1) How many internal vertices are in a full 5-ary tree with 101 leaves?

   A. 30 B. 33 C. 25 D. 28

13. (L.O.3.1) A grocery store has a promotion for customers when shopping. The promotional gift is any 10 toothbrushes from Colgate, Oral-B, and P/S brands. How many different ways to create promotional gifts?

   A. 66 B. 23 C. 87 D. 12

14. (L.O.2.2) How many binary strings of length 8 do not contain at least six consecutive 1s?

   A. 246. B. 248. C. 249. D. 250.

15. (L.O.2.2) Given a W_3 graph, the cycle vertices are indexed from 1 to 3, and the middle vertex is 4. The value of each edge is the sum of its adjacency vertex. For instance, the edge between vertex 2 and 3 has a value of 5. What is the expectation of the total value of a spanning tree chosen from this W_3?

A. 16 B. 15 C. 15.5 D. 14.25

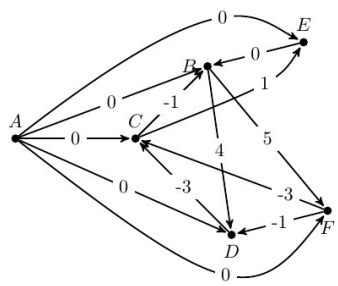
16. (L.O.3.1) An athlete decides to climb the mountain on a specific day from A to B. If this person has an **accident** or **bad weather**, he will immediately stop climbing and return to A. According to the survey this season, the possibility of a day with good weather is 60%, normal weather is 30%, and bad weather is 10%. Know that the chance of this athlete having an accident in good weather is 1%, and this chance is 5% in normal weather. Find the probability that this athlete reaches B.

A. 87.9%. B. 2.1%. C. 97.9%. D. 12.1%.

17. (L.O.2.2) Consider a team chess match, knowing that each team has 5 players, and each player will have to play games against every player from the opponent team. The team that has more wins in the match is the winning team, and the probability of winning for each player is the same. What kind of graph represents the match, and what is the probability of a team winning?

A. $K_{5,5}$, 0.5 B. All the others answers are incorrect
C. $W_{5.5}$, 0.422 D. $K_{5,5}$, 0.589

18. (L.O.1.2) Given the following graph,



What is the shortest path (total weight) from A to the remaining vertices B, C, D, E, F:

A. Another answer. B. $-4, -5, 1, -1, 0$. C. $-5, -4, -1, -3, 0$. D. $-4, -4, -1, -2, 0$.

19. (L.O.2.2) A building has a ground floor G and other four floors marked 1,2,3,4. Seven people get in the elevator at G. How many ways can they get out of the elevator on the remaining floors if

(a) every floor has at least one person get out and,

(b) the order of the people coming out on the same floors does not matter

A. 1913. B. 8400. C. 1319. D. 4800.

20. (L.O.2.2) Given a discrete structure final test with 5 questions on Counting, 5 questions on Probability, 4 questions on Graph, 4 questions on Connectivity, and 7 on Tree. A sample student who has 30% of doing a Counting question right, 40% of doing a Probability question right, 50% of doing a Graph question right, 60% of doing a Connectivity question wrong, and 70% of doing a Tree question right. What is the expected number of correct questions this student can have?

A. 11 B. 12 C. 12.5 D. 13

21. (L.O.2.2) The number of relations that are both reflexive and symmetric on a set of 2023 elements is

A. $2^{\frac{2022*2023}{2}}$. B. 2^{2023^2}. C. $2^{\frac{2022*2023}{2}+1}$. D. $2^{2022*2023}$.

22. (L.O.1.2) Let's consider an undirected graph given by the following adjacency list.

Vertex	Adjacent vertices
a	b, d, f
b	a, c, e
c	b, d, h
d	a, c, g
e	b, f, h
f	a, e, g
g	d, f, h
h	c, e, g

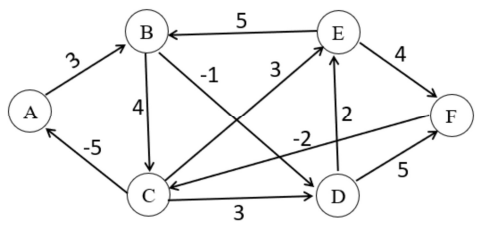
Which of the following assessments is correct?

A. All answers are correct.
B. The graph is bipartite.
C. The graph has a Hamilton circuit.
D. The graph has no Euler circuit.

23. (L.O.1.2) The degree sequence of a simple graph is the sequence of the degrees of the nodes in the graph in decreasing order. Which of the following sequences can not be the degree sequence of any graph?
(I) 7,6,5,5,5,4,4,2,2; (II)7,6,6,6,4,4,2,2; (III) 8,7,7,6,4,2,2,1,1; (IV)9,9,6,6,6,6,5,4,3,2

A. II and IV.
B. I and IV.
C. I only.
D. II and III.

24. (L.O.3.2) Consider the weighted directed graph G



The total weights of the edges of the shortest paths from vertex A to vertices B, C, D, E, and F, respectively, are

A. The shortest paths do not exist.
B. 3, 5, 2, 4, and 7.
C. 3, 7, 2, 4, and 7.
D. 3, 7, 4, 5, and 7.

25. (L.O.1.2) Find in-order traversal of a binary tree with pre-order $ADEBJCFHIG$ and post-order $EJBDHIFGCA$

A. $ADJBEHFICG$
B. $EDJBAHFICG$
C. $EDJBAHFCIG$
D. $EDJBAFHICG$

Solution 2225

1. D.
2. B.
3. B.
4. B.
5. B.
6. B.
7. D.
8. A.
9. B.
10. D.
11. B.
12. C.
13. A.
14. B.
15. B.
16. A.
17. B.
18. C.
19. B.
20. B.
21. A.
22. A.
23. D.
24. B.
25. B.